DVĪPAHAṂSA

ISLAND HOPPER AMPHIBIAN AIRCRAFT

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Introduction

This project presents the conceptual design of the Dvīpahaṃsa, a regional amphibious aircraft that carries 17 passengers and three crew over a design range of about 1550 km. It is built for inter-atoll "island hopping" in places like the Maldives, where paved runways are scarce and helicopters are limited in range and payload. Sized at a maximum takeoff weight of 7,220 kg, the design aims for efficient cruise at 85 m/s, water-takeoff performance and sound stability across its loading range.

Design Requirements

- Takeoff and landing on both water and land (amphibious operation).
- Carry 17 passengers and 3 crew members, with 20 kg of baggage per passenger.
- Water takeoff run of 900 m or less.
- Stall speed within 77.75 knots (40 m/s).
- Cruise speed of 165 knots (85 m/s) at a cruising altitude of 9,000 ft.
- Design range of more than 1000 Km.
- Statically and dynamically stable aircraft with proper handling qualities.

Aircraft Configuration

Conventional Hull (V-Bottom Ship-Hull Fuselage)

- Gives the best hydrodynamic performance for takeoff and landing on open water.
- Works with existing water bases and airstrips, so it needs no special infrastructure.
- Wheeled landing gear and wingtip floats allow land operation and keep the aircraft stable on water.

High Monoplane Wing

- Provides maximum clearance above the water for the engines and propellers.
- Gives better aerodynamic efficiency at cruise and a lighter, simpler structure than a biplane.

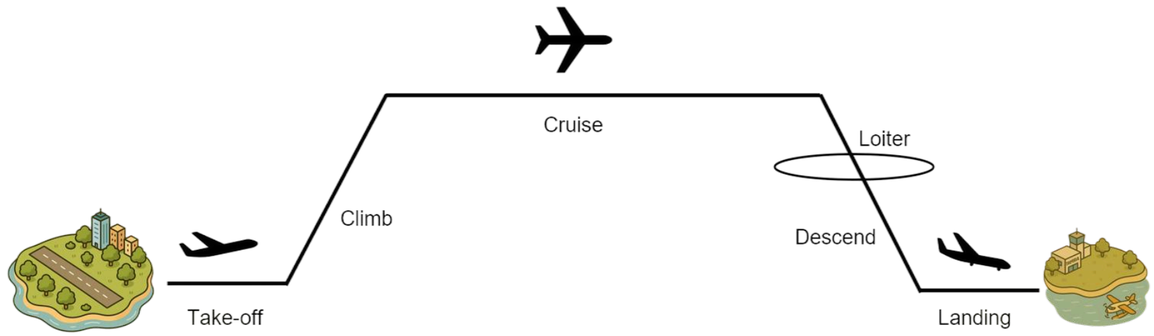
T-Tail

- Puts the horizontal stabilizer above the water spray and out of the wing wake.
- Improves pitch stability and tail effectiveness and lowers the risk of spray damage to the tail.

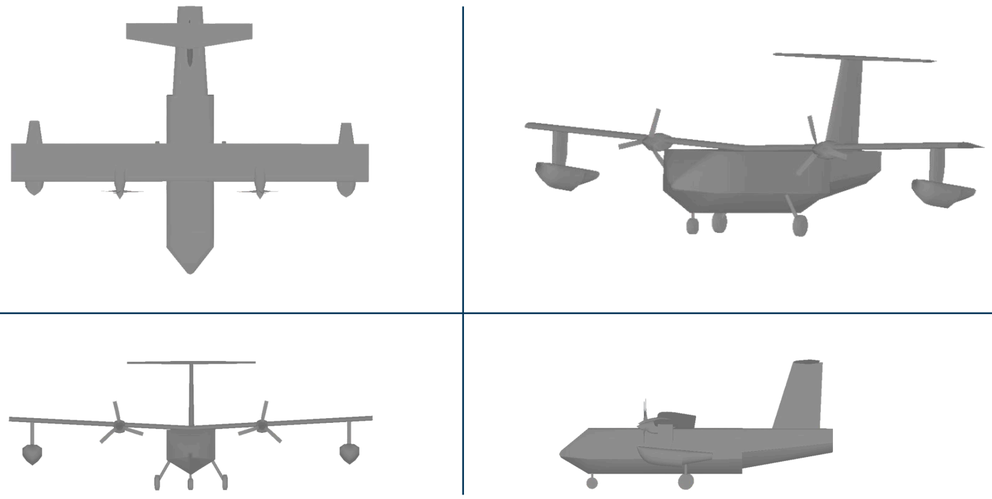
Twin Turboprop (Tractor) Engines

- Gives the efficiency and power needed for short-field and water takeoff over island-hopping ranges.
- The wing-mounted nacelles sit high above the water line.

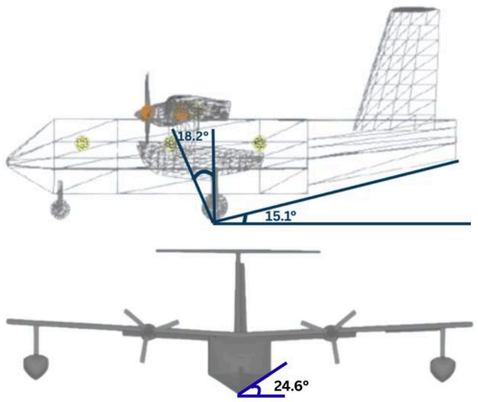
Mission Profile



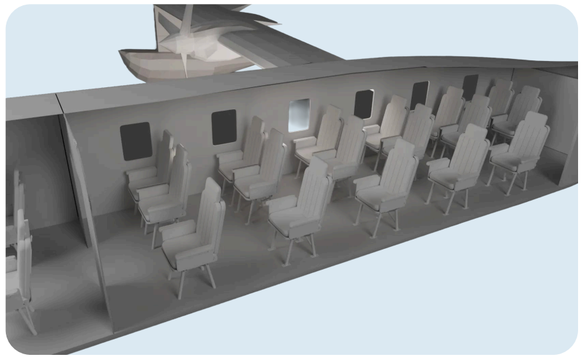
Aircraft Layout



Tipback Angle & Deadrise Angle



Interior Layout



Specification

PARAMETERS	VALUES
MTOW	7220 kg
Range	1550 km
Endurance	5 hrs
Stall Speed	78 kts
Cruise Speed	165 kts
Maximum Speed	198 kts
Cruising Altitude	9000 ft
Payload Weight	1954 kg
Empty Weight	4254 kg
Fuel Capacity	1012 kg
Power	1325 hp
Fuselage Length	15.9 meters

Wing & Tail Sizing

SURFACE	WING	HORIZONTAL TAIL	VERTICAL TAIL
Airfoil	NACA 4412	NACA 0012	NACA 0012
Span (m)	18.9	7.86	3.52
Area (m^2)	38	9.86	8.24
AR	9.4	6.27	1.85
Incidence	0°	−2°	0°
Taper Ratio	1	0.8	0.6

Control Surface Sizing

Surface	% chord	% span	Deflection
Aileron	25	40	−15° to 25°
Elevator	40	80	−30° to 20°
Rudder	40	90	±30°
Flap	20	40	0° to 25°

Matching Plot

